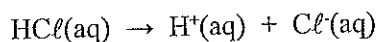


Problem-solving and quantities in chemistry

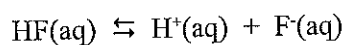
Set 12: Acid and base strength

Strength of acids and bases refers to the degree of ion formation that occurs when an acid or base is dissolved in water as a proportion of the solute actually dissolved.

A **strong acid**: When hydrogen chloride dissolves in water it ionises completely to form ions (hydrochloric acid). The ionisation process is represented by the equation:



A **weak acid**: When the very soluble hydrogen fluoride dissolves in water there is little ionisation. The ionisation process is represented by the equation:



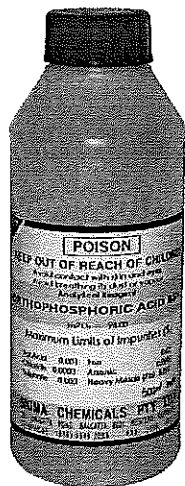
Concentration is a description or measure of the proportion of solute in a solution. It can refer to the original form of the solute or to the resulting species formed as a result of the solution process. For example a 6 mol L^{-1} solution is concentrated while a 0.1 mol L^{-1} solution is dilute.

Set 12: Exercises

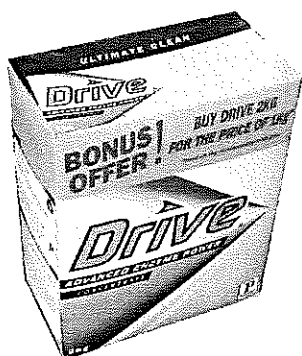
1. People often talk about strong or weak solutions when they actually mean concentrated or dilute solutions. With the aid of diagrams and equations distinguish between the properties of 'concentration' and 'strength' when applied to solutions of acids and bases.
2. The uses of acids and bases are to a large extent determined by their strength. Write three examples of:
 - (a) Strong acids
 - (b) Weak acids
 - (c) Strong bases
 - (d) Weak bases
3. Classify each of the following solutions on the basis of concentration and strength.
 - (a) 10.0 mol L^{-1} hydrofluoric acid solution used in etching glass.
 - (b) $0.0100 \text{ mol L}^{-1}$ hydrochloric acid solution used in the analysis of basic solutions.
 - (c) 10.0 mol L^{-1} sodium hydroxide solution used in the purification of alumina.
 - (d) 0.100 mol L^{-1} ammonia solution used as a component of some floor cleaners.
4.
 - (a) Write an equation or equations to illustrate the process that occurs when 0.1 mol of each of the following are mixed with 1.0 L of water.
 - (b) In each case indicate whether the process is ionisation or dissociation?
 - (c) In each case write the formula of all the species present in order of increasing concentration.
 - (i) hydrogen bromide
 - (ii) acetic acid
 - (iii) sulfuric acid
 - (iv) ammonia
 - (v) barium hydroxide

Notes

Set 12: Acid and base strength



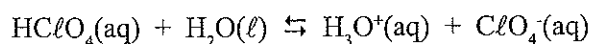
Question 7: Concentrated phosphoric acid



Question 9: Washing Powder

Notes

5. Consider the equation:



It represents the reaction of perchloric acid with water to produce hydronium and perchlorate ions. Salts that contain the perchlorate ions are increasingly used to make explosive mixtures used in fireworks. It was found that such a solution contained only a very small number of perchloric acid molecules. Which is the stronger acid and which is the weaker base present in the solution.

6. Hydrogen sulfide is produced by certain anaerobic organisms that use sulfur compounds for respiration instead of oxygen. It is readily soluble in water.
- Write equations to represent the process that occurs when hydrogen sulfide has dissolved in water.
 - Write the formulae of the strongest acid and the strongest base present in the resulting solution.
7. Concentrated phosphoric acid is used to convert phosphate rock into a fertiliser known as triple superphosphate.
- Write equations to show what happens when about 10 mL of phosphoric acid is dissolved in 1.0 L of water.
 - Identify all the molecules and ions present in the solution and write them in order of decreasing concentration.
8. With precautions, concentrated sulfuric acid (99 %) can be safely stored in iron tanks but a 2 mol L⁻¹ sulfuric acid solution readily dissolves steel wool. Account for these observations.
9. Hydroxide ions in solution readily react with fats and oils to produce salts that are relatively water soluble. Both sodium carbonate and sodium hydroxide produce hydroxide ions when dissolved in water. Washing powders used to wash clothing contains amongst other things sodium carbonate. Why is sodium carbonate preferred to sodium hydroxide for this use?
10. One of the uses of boric acid, H₃BO₃, is as an antiseptic and to treat certain yeast infections in humans. Small amounts are even used in eye wash solutions. Based on these uses comment on the likely strength of this acid. Why would it not be appropriate to use hydrochloric acid for these uses?

Set 13: Hydrolysis

Hydrolysis means reaction with water. When ions associated with weak acids or bases are dissolved in water they undergo hydrolysis, they react with water. It is a chemical reaction where one or more water molecules are split into hydrogen and hydroxide ions, which may participate in further reactions. Dissolved salts in soils can make the soil either acidic or basic.

Table 13.1: Acid and base properties of common aqueous anions and cations

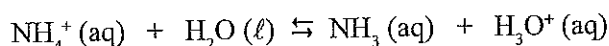
Acidic	Neutral	Basic
Anions from polyprotic acids: HSO_4^- , H_2PO_4^-	Anions from strong acids: NO_3^- , Cl^- , Br^- , I^-	Anions from weak acids: CH_3COO^- , CO_3^{2-} , HCO_3^- , PO_4^{3-} , HPO_4^{2-} , ClO^- , F^- , S^{2-} , SO_4^{2-}
Acidic cations: NH_4^+ , Al^{3+} , Fe^{3+}	Cations from strong bases: Li^+ , Na^+ , K^+ , Mg^{2+} , Ca^{2+} , Ba^{2+}	

Example

When the strong acid HCl reacts with the weak base NH_3 the acidic salt NH_4Cl is formed.



The ammonium ion, NH_4^+ , hydrolyses in water according to the equation:

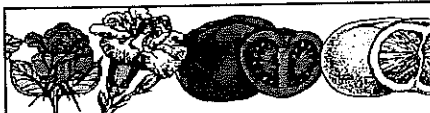


NH_4Cl is an acidic salt because the NH_4^+ ion reacts with water to produce hydronium ions, H_3O^+ . The Cl^- does not hydrolyse.

Set 13: Exercises

- Classify the following salts as acidic, basic or neutral.
 - sodium chloride
 - potassium phosphate
 - calcium hydrogencarbonate
 - ammonium nitrate
 - sodium acetate
 - aluminium hydrogensulphate
- Write hydrolysis equations for the acidic and basic salts identified in question 1.
- Many salts that contain essential elements for plant growth are used as fertilisers. In common use are salts such as ammonium nitrate, sodium nitrate, calcium hydrogenphosphate, potassium chloride, potassium sulfate and ammonium chloride.
 - Which of these salts have the potential to change the pH of the soils in which they are used?
 - Explain how you decided which of these would change the soil's pH?
 - Write equations that show how the soil's pH would be increased or decreased by these salts.
 - Why do some salts used as fertilisers not change the soil's pH?

Notes



Nitrogen (N)	12.0%	Sulphur (S)	7.0%
Phosphorus (P)	2.6%	Calcium (Ca)	3.5%
As Water Soluble	2.6%	Trace Elements	
As Citrate Soluble	5.2%	Magnesium (Mg)	1.2%
Total		Iron (Fe)	0.1%
Potassium (K)	14.1%	Zinc (Zn)	0.01%
As Sulphate		Boron (Bo)	0.02%
Total Potassium	14.1%		

CONCENTRATED FERTILISER

Question 3: Fertiliser

Set 13: Hydrolysis

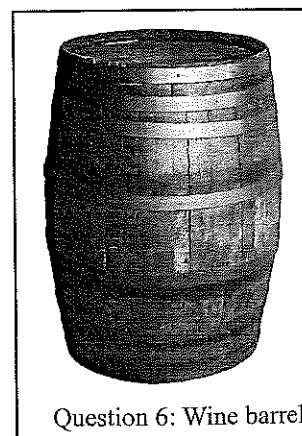
Notes

4. Calcium dihydrogenphosphate, which is sparingly soluble, is the main compound in the fertiliser superphosphate. It is a source of the element phosphorus, essential for plant growth. Write an equation to show what happens when calcium dihydrogenphosphate dissolves in water.
5. Washing soda (sodium carbonate decahydrate) is a major component of washing powders.

INGREDIENTS

DRIVE contains ingredients to lift dirt from clothes (anionic and nonionic surfactants); soften water (sodium polyphosphate and zeolite); break up fatty soils (sodium carbonate, sodium silicate) and remove stains (enzymes). Small amounts of fluorescer, soil responding agent, colour and perfume are also included.

- (a) When dissolved in water, does the washing soda change the pH of the water? If so, does the solution become acidic or basic?
- (b) Write an equation to represent the solution process and any other equations that may be required to explain your answer to part (a).
6. Used wine barrels can sometimes end up with acid forming in them as the alcohol in any residual wine is converted to acetic acid. Before they can be used to store wine again, this acetic acid must be removed. This can be achieved by washing the barrel with a dilute solution of sodium hydroxide.
- (a) Write an equation for the reaction between the acetic acid and sodium hydroxide solutions.
- (b) If the salt formed from this reaction was recovered, dried then dissolved in distilled water would the pH of the solution formed be greater than 7, equal to 7 or less than 7?
- (c) Write an equation to justify your answer to part (b).
7. Sodium fluoride is added in very small amounts (0.7 to 1.2 ppm) to our drinking water to reduce tooth decay as the tooth enamel that contains some fluorine atoms seems to be much more resistant to attack from acids produced by mouth bacteria than normal enamel. Does the addition of sodium fluoride to water change its pH? Write an equation or equations to support your answer.
8. Ammonium acetate could be used as a fertiliser as it contains the element nitrogen that is important for plant growth. Would you expect a solution of ammonium acetate to be acidic, neutral or basic? You may need to do some research to find the answer. Write equations to support your answer.



Question 6: Wine barrel

Set 14: Water equilibrium

Ionisation Constant of Water (K_w)

Water is a weak electrolyte and dissociates to a very small extent according to the equation.



The equilibrium constant for this reaction at 25 °C is:

$$K_w = [\text{H}^+][\text{OH}^-] = 1.00 \times 10^{-14}$$

where $[\text{H}^+]$ is the concentration of hydrogen ions in mol L^{-1} , and $[\text{OH}^-]$ is the concentration of hydroxide ions in mol L^{-1} .

Examples

- Hydrobromic acid is a strong acid and is used for making brominated hydrocarbons. Calculate the hydrogen and hydroxide ion concentration of a $2.50 \times 10^{-3} \text{ mol L}^{-1}$ aqueous solution of hydrobromic acid.

(a) Write an ionisation equation:



(b) Calculate the hydrogen ion concentration:

As HBr is a strong acid, it is fully ionised.

So the:

$$[\text{H}^+] = [\text{HBr}] = 2.50 \times 10^{-3} \text{ mol L}^{-1}$$

(c) Calculate the hydroxide ion concentration:

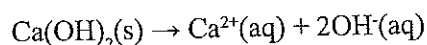
Since:

$$K_w = [\text{H}^+][\text{OH}^-] = 1.00 \times 10^{-14}$$

$$[\text{OH}^-] = \frac{1.00 \times 10^{-14}}{[\text{H}^+]} = \frac{1.00 \times 10^{-14}}{2.50 \times 10^{-3}} = 4.00 \times 10^{-12} \text{ mol L}^{-1}$$

- Calcium hydroxide is sparingly soluble in water and at 25°C its concentration is around $0.0158 \text{ mol L}^{-1}$. Calculate the concentration of the hydroxide and hydrogen ion for this solution.

Step 1: Write a dissociation equation:



As Ca(OH)_2 is ionic it is fully dissociated.

Notes

Set 14: Water equilibrium

Notes

Step 2: Calculate the hydroxide ion concentration:

$$[\text{OH}^-] = 2 \times [\text{Ca}(\text{OH})_2] = (2) (0.0158) = 0.0316 \text{ mol L}^{-1}$$

Since:

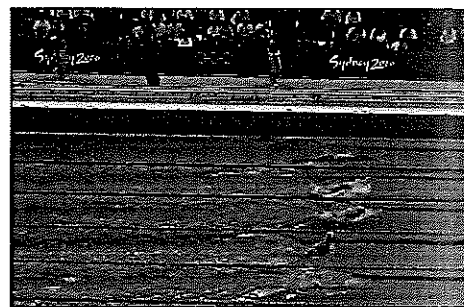
$$K_w = [\text{H}^+][\text{OH}^-] = 1.00 \times 10^{-14}$$

$$[\text{H}^+] = \frac{1.00 \times 10^{-14}}{[\text{OH}^-]} = \frac{1.00 \times 10^{-14}}{0.0316} = 3.16 \times 10^{-13} \text{ mol L}^{-1}$$

Set 14: Exercises

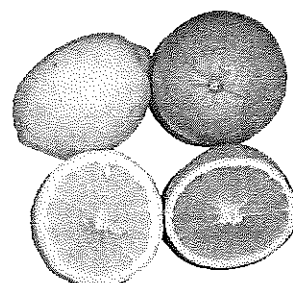
In all of the following questions the temperature is assumed to be 25 °C

- Write an equation to represent the ionisation of water according to the Brønsted-Lowry theory of acids and bases.
- The concentration of acid and base in a swimming pool is critical to maintaining an adequate level of chlorine in the water. The concentration of hydrogen ions in the water of an olympic swimming pool was measured to be $1.48 \times 10^{-7} \text{ mol L}^{-1}$.
 - Calculate the concentration of hydroxide ions.
 - Is the swimming pool water basic or acidic?
- Water in the tailings dam of a gold processing plant was found to have a hydroxide ion concentration of $1.58 \times 10^{-6} \text{ mol L}^{-1}$. Calculate the $[\text{H}^+]$.
- Calculate the hydrogen ion concentration in the following solutions.
 - A standard $1.55 \times 10^{-4} \text{ mol L}^{-1}$ sodium hydroxide solution used for analysing the acid content of wine.
 - A $3.90 \times 10^{-2} \text{ mol L}^{-1}$ barium hydroxide solution used to make soap.
- Calculate the hydroxide ion concentration in the following solutions.
 - A standard 0.104 mol L^{-1} hydrochloric acid used to analyse floor cleaning solutions
 - A 0.125 mol L^{-1} sulfuric acid solution used to dissolve samples of malachite (copper ore containing mostly copper carbonate). Assume the first hydrogen is fully ionised and the second is 10.0% ionised.
- The ionisation constant for water is usually taken to be 1.00×10^{-14} at 25 °C.
 - Why is a temperature stated?
 - What effect you would observe on the equilibrium constant if the temperature of pure water was decreased. Explain in terms of Le Chatelier's principle.
 - What would be the effect on the hydrogen ion concentration in pure water if the temperature was increased above 25 °C?
 - Explain this effect in terms of changes in reaction rates.



Question 2: Swimming pool

7. Rain water collected near a nickel smelter processing nickel sulfide ore was found to have a hydrogen ion concentration of $1.55 \times 10^{-5} \text{ mol L}^{-1}$.
- Calculate the $[\text{OH}^-]$.
 - What could have caused the rain water to have such a high $[\text{H}^+]$?
 - Write an equation or equations to show how the $[\text{H}^+]$ in the rain water could become greater than $1.00 \times 10^{-7} \text{ mol L}^{-1}$.
8. Lemon juice with a $[\text{H}^+]$ of $1.60 \times 10^{-4} \text{ mol L}^{-1}$ was mixed with an equal volume of orange juice with a $[\text{H}^+]$ of $1.30 \times 10^{-6} \text{ mol L}^{-1}$ to make a citrus juice drink. Calculate the hydrogen and hydroxide ion concentration of the citrus juice drink.
9. When you suffer from an acid stomach, the fluid in your stomach can contain hydrochloric acid at a concentration of $2.50 \times 10^{-4} \text{ mol L}^{-1}$.
- To reduce the amount of acid in your stomach you are advised to take an antacid tablet that contains 5.00 mg of aluminum hydroxide and 5.00 mg of magnesium hydroxide. If at this time you have 1.50 L of fluid in your stomach, calculate the concentration of hydrogen ions and hydroxide ions in your stomach after you have taken the tablet.
 - If you took a second antacid tablet what would the hydrogen and hydroxide ion concentration be now?
10. Your friend complains to you of a burning sensation in their stomach just before a chemistry test. You both suspect that this is caused by hyperacidity, an unusually high amount of acid in the stomach, so your friend decides to seek medication from the resident nurse. The nurse decides to use milk of magnesia, an 80 g L^{-1} slurry of solid magnesium hydroxide suspended in water, to reduce the amount of acid in your friend's stomach. The nurse estimated that your friend's stomach contained 2.00×10^{-2} moles of hydrochloric acid.
- Calculate the volume of milk of magnesia the nurse should administer to reduce the hydrogen ion concentration in your friend's stomach to one third of the original value.
 - Calculate the hydrogen ion and hydroxide ion concentration in your friend's stomach after he took the medication if the volume of the fluid in his stomach was estimated to be 800 mL.



Question 8: Lemons and oranges

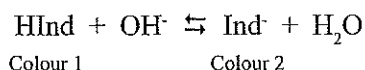
Notes

Set 15: Indicators and their use

Notes

Acid-base indicators are dyes whose colours depend upon the hydrogen ion concentration in solution. Each indicator changes colour at a unique $[H^+]$.

The neutralisation reaction of an acid base indicator can be represented by the equation:



The colour of the indicator in a solution depends on the relative concentration of $HInd$ (acid) and Ind^- (conjugate base). The colour change for different indicators occurs at different pH values. Using the equation as an example, colour 1 would be observed at a lower pH value than colour 2. If the pH were to be increased the indicator, over a specific pH range, would change to colour 2. Table 15.1 shows some acid-base indicators and their colours at different pH values.

Table 15.1 Acid-base indicators and their colour change

Indicator	Colour on acid side	pH range of colour change	Colour on base side
Methyl violet	yellow	0.0 – 1.6	violet
Bromophenol blue	yellow	3.0 – 4.6	blue
Methyl orange	red	3.1 – 4.4	yellow
Methyl red	red	4.4 – 6.2	yellow
Litmus	red	5.0 – 8.0	blue
Bromothymol blue	yellow	6.0 – 7.6	blue
Phenolphthalein	colourless	8.3 – 10.0	pink
Alizarin yellow	yellow	10.1 – 12.0	red
Indigo carmine	blue	11.4 – 13.0	yellow

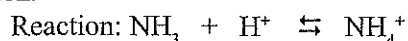
Indicators are chosen and used to show the pH of aqueous solutions such as swimming pool water and most importantly in volumetric analysis **titrations**. Indicators are selected so that their colour change occurs over the pH range that is important for the task that they are being used for.

Titration

The process of adding one solution to another until reaction between them is complete is known as titration. One solution, the standard, contains a known quantity of one reactant and a titration can be used to find out information about the other solution from knowledge of the reaction involved. When chemically equivalent amounts of acid and base have been added the equivalence point has been reached. Indicators are used to determine the equivalence point of a reaction. It does this by changing colour indicating the end point. The indicator is chosen so that its colour change, that is the end point, matches as closely as possible the equivalence point of the titration reaction.

Example

Which indicator would you use in the titration of a cloudy ammonia solution (NH_3) and hydrochloric acid solution.



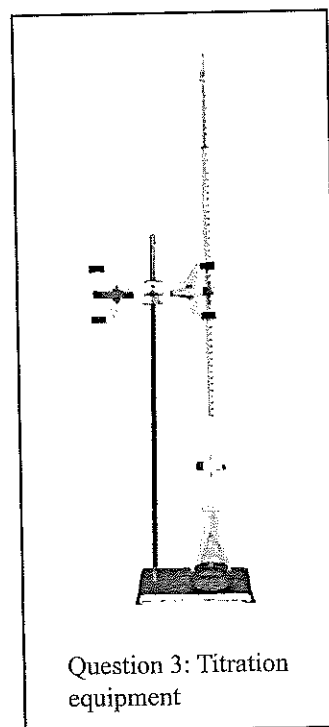
At the equivalence point of the reaction the acidic cation NH_4^+ and the neutral anion Cl^- are present (see table 13.1). I would choose an indicator that changes colour at an acidic pH such as methyl red (see table 15.1).

Set 15: Exercises

- Knowledge of the acidity or alkalinity of swimming pool water is vital for proper maintenance of the pool and associated equipment and for the comfort of swimmers. Use of acid – base indicators is a relatively simple method of checking the pH of the water.
 - What type of compound is usually used as an acid – base indicator?
 - Describe the essential properties that a compound must possess to be a useful indicator.
- Write an equation to show what happens when an indicator is dissolved in water.
 - Describe how the addition of an acid results in a change of colour of the solution.
 - Describe how the addition of a base results in a change of colour of the solution.
 - Explain in terms of changes in reaction rates why these colour changes occur.
- Indicators are often used to detect the end point during the analysis of acids or bases. During the analysis of fruit juice the end point is known to occur when the reaction mixture is alkaline.
 - Give an example of an indicator that will change colour in an alkaline solution.
 - What are the colours that this indicator can have at various levels of acidity?
 - What is the pH range where the indicator changes colour?
- During the analysis of formalin solutions that contain the weak base methanal, the end point is known to occur when the reaction mixture is acidic.
 - Give an example of an indicator that will change colour in an acid solution.
 - What are the colours that this indicator can have at various levels of acidity?
 - What is the pH range where the indicator changes colour?
- The active ingredient of vinegar is the organic acid, acetic acid. A Consumer Affairs agency was investigating complaints that a particular brand of vinegar was not the advertised concentration. A chemist was commissioned to determine its concentration. Volumetric analysis using standard sodium hydroxide solution was the method chosen by the chemist. Determine the following for the chemist.
 - What species are present in the reaction mixture at the equivalence point? List them in order of decreasing concentration.
 - Sketch a graph of the change in pH that occurs in the reaction mixture as the acetic acid is added to the sodium hydroxide solution. Continue the graph until a significant excess of the vinegar has been added to the base in the flask.
 - At equivalence point is the solution acidic, neutral or basic? Write an equation to justify your answer.
 - Choose an appropriate indicator to determine the end point. Explain your choice.



Question 1: pH test kit



Question 3: Titration equipment

Notes

Set 15: Indicators and their use

Notes

6. One of the components of floor cleaners is ammonia. To make sure that the floor cleaner contains the correct concentration of ammonia the quality control chemist conducted an analysis of a sample of floor cleaner. Volumetric analysis using standard hydrochloric acid solution was the method used. Determine the following for the chemist.
- (a) What species are present in the reaction mixture at the equivalence point? List them in order of decreasing concentration.
 - (b) Sketch a graph of the change in pH that occurs in the reaction mixture as the hydrochloric acid is added to the floor cleaner solution. Continue the graph until a significant excess of the acid has been added to the base in the flask.
 - (c) At equivalence point is the solution acidic, neutral or basic? Write an equation to justify your answer.
 - (d) Choose an appropriate indicator to determine the end point. Explain your choice.
7. Builders' lime is often supplied as a thick slurry of calcium hydroxide suspended in water. When left in a container for some time some of the solid settles leaving a clear saturated calcium hydroxide solution above the solid. You were asked to determine the concentration of this solution using volumetric analysis. Standard hydrochloric acid solution was the suggested reagent for the analysis. Determine the following.
- (a) What species are present in the reaction mixture at the equivalence point? List them in order of decreasing concentration.
 - (b) Sketch a graph of the change in pH that occurs in the reaction mixture as the hydrochloric acid is added to the calcium hydroxide solution. Continue the graph until a significant excess of the acid has been added to the base in the flask.
 - (c) At equivalence point is the solution acidic, neutral or basic? If possible write an equation to justify your answer.
 - (d) Choose an appropriate indicator to determine the end point. Explain your choice.
8. Oxalic acid is a constituent of edible plants such as rhubarb and spinach. You are asked to analyse the oxalic acid content of these plants by titration. During the analysis the oxalic acid solution is added to a standard sodium hydroxide solution. Methyl orange and phenolphthalein are the only two indicators available to you.
- (a) Which indicator should you use to obtain an accurate analysis?
 - (b) If you chose the other indicator, would you use more or less oxalic acid solution to reach the end point?
 - (c) How would using the wrong indicator affect the value you obtained for the oxalic acid content of the plants you measured?



Question 8: Rhubarb

Set 16: The pH scale

The pH scale is a convenient way of expressing concentrations of hydrogen ions, $[H^+]$ in aqueous solutions. The definition of pH is expressed by the equation:

$$pH = -\log_{10} [H^+] \text{ where pH is a number without units.}$$

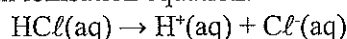
The pH scale has a usual range of 0 to 14 as illustrated in the diagram below:

pH	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
$[H^+]$	$1=10^0$	10^{-1}	10^{-2}	10^{-3}	10^{-4}	10^{-5}	10^{-6}	10^{-7}	10^{-8}	10^{-9}	10^{-10}	10^{-11}	10^{-12}	10^{-13}	10^{-14}
Acidic pH < 7								Neutral pH = 7	Basic pH > 7						

Examples

- Hydrochloric acid is one of the most common acids with many uses. It is produced in Western Australia in large quantities as a by product of the refining of titanium dioxide from the mineral sand ilmenite. Low concentration solutions of the acid can be standardised and used in analysis of basic solutions. Calculate the $[OH^-]$, $[H^+]$ and pH of a $0.0200 \text{ mol L}^{-1} \text{ HCl}$ solution

(a) Write an ionisation equation:



(b) Calculate the hydrogen ion concentration:

As HCl is a strong acid, it is fully ionised.

$$\text{So the } [H^+] = 0.0200 \text{ mol L}^{-1}$$

(c) Calculate the hydroxide ion concentration:

Since:

$$K_w = [H^+][OH^-] = 1.00 \times 10^{-14}$$

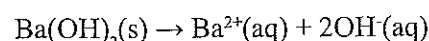
$$[OH^-] = \frac{1.00 \times 10^{-14}}{[H^+]} = \frac{1.00 \times 10^{-14}}{0.0200} = 5.00 \times 10^{-13} \text{ mol L}^{-1}$$

(d) Calculate the pH of the solution:

$$pH = -\log_{10} [H^+] = \log_{10} (0.0200) = 1.70$$

- Barium hydroxide is one of the few soluble metal hydroxides. It can be used in the analysis of acidic solutions. Calculate the $[H^+]$, $[OH^-]$ and pH of a $3.00 \times 10^{-3} \text{ mol L}^{-1} \text{ Ba(OH)}_2$ solution.

(a) Write a dissociation equation:



Notes

Set 16: The pH scale

Notes

- (b) Calculate the hydroxide ion concentration:
As $\text{Ba}(\text{OH})_2$ is ionic it is fully dissociated.

$$[\text{OH}^-] = 2 \times [\text{Ba}(\text{OH})_2] = (2) (3.00 \times 10^{-3}) = 6.00 \times 10^{-3} \text{ mol L}^{-1}$$

- (c) Calculate the hydrogen ion concentration:
Since:

$$K_w = [\text{H}^+][\text{OH}^-] = 1.00 \times 10^{-14}$$

$$[\text{H}^+] = \frac{1.00 \times 10^{-14}}{[\text{OH}^-]} = \frac{1.00 \times 10^{-14}}{6.00 \times 10^{-3}} = 1.67 \times 10^{-12} \text{ mol L}^{-1}$$

- (d) Calculate the pH of the solution:

$$\text{pH} = -\log_{10} [\text{H}^+] = \log_{10} (1.67 \times 10^{-12}) = 11.8$$



3. Orange juice has a pH of 4.50. What are the $[\text{H}^+]$ and $[\text{OH}^-]$ concentrations in orange juice?

- (a) Calculate the hydrogen ion concentration:

$$\text{pH} = -\log_{10} [\text{H}^+]$$

$$-4.50 = \log_{10} [\text{H}^+]$$

$$[\text{H}^+] = 10^{-4.50} = 3.16 \times 10^{-5} \text{ mol L}^{-1}$$

- (b) Calculate the hydroxide ion concentration:
Since:

$$K_w = [\text{H}^+][\text{OH}^-] = 1.00 \times 10^{-14}$$

$$[\text{OH}^-] = \frac{1.00 \times 10^{-14}}{[\text{H}^+]} = \frac{1.00 \times 10^{-14}}{3.16 \times 10^{-5}} = 3.16 \times 10^{-10} \text{ mol L}^{-1}$$

Set 16: Exercises

In all of the following questions the temperature is assumed to be 25 °C

1. Calculate the $[\text{H}^+]$, $[\text{OH}^-]$ and pH of the following solutions.
 - (a) A 0.100 mol L⁻¹ standard HCl solution used to analyse ammonia based floor cleaners.
 - (b) A 0.00500 mol L⁻¹ HNO₃ solution used to reduce the pH of nutrient solutions used for hydroponics.
 - (c) A 0.0100 mol L⁻¹ standard NaOH solution used to analyse vinegar solutions.
 - (d) A 2.00 mol L⁻¹ HCl solution used to dissolve limestone prior to analysis.
 - (e) A 2.45 mol L⁻¹ NaCl solution used to preserve olives.

2. Calculate the $[H^+]$ and $[OH^-]$ of solutions with the following pH:

- (a) Lemon juice of pH 3.00
- (b) Dish washing solution of pH 11.0 made from dishwashing powder used in a dishwasher.
- (c) Pool acid of pH -1.00
- (d) Orange juice of pH 4.56
- (e) Swimming pool water of pH 7.60

3. Before disposal, a hydrochloric acid solution of pH 4.00 used to remove rust from iron parts prior to galvanising needs to be neutralised. Base is added until the pH is 7.00. By what factor has the hydrogen ion concentration changed?

4. A brick cleaner needs to replenish the hydrochloric acid solution he is using. Using a pH meter he measures the pH of his depleted solution to be 2.00. He decides to add 3.00 L of 3.00 mol L⁻¹ HCl solution to 2.00 L of his solution. Calculate the final hydrogen ion concentration of the new solution.

5. A laboratory chemist wants a low concentration hydrochloric acid solution with a pH of 5.00, for calibrating a conductivity meter. What volume of water must be added to a 25.0 mL sample of hydrochloric acid solution of pH 3.60 to produce the required solution?

6. A researcher studying the extraction of alumina from bauxite requires 100 mL of a sodium hydroxide solution of pH 12.0. When the stock solution was checked it was found to have a pH of 11.7. What mass of sodium hydroxide must be added to increase the pH of this solution to 12.0? Assume the volume of the solution does not change with the addition of the solid.

7. An environmental chemist is conducting research into highly alkaline solutions found in a drain near a concrete batching plant. The concentration of the solution was found to be 0.236 mol L⁻¹ hydroxide ion and the runoff from the plant was found to have a concentration of hydroxide ion of 0.156 mol L⁻¹. The chemist mixed 200 mL of the solution from the drain with 300 mL of the runoff from the plant.

- (a) Calculate the pH of the resulting solution.
- (b) What volume of 1.00 mol L⁻¹ hydrochloric acid solution does the chemist need to add per litre of the mixed solution to neutralise it.

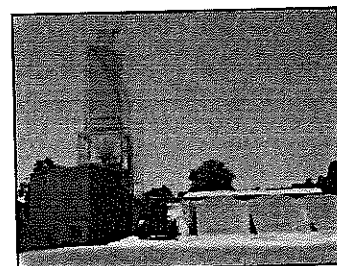
For both (a) and (b) Assume the volumes are additive.

8. A water tank contains 15 000 L of bore water of pH 5.50. In an effort to increase the pH 10.0 g of sodium hydroxide was added to the water in the tank. Determine the pH of the resulting solution. Assume the volume does not change on adding the solid.

9. A swimming pool contains 2.00 ML of water at a pH of 7.80 making swimming in it unpleasant. The caretaker needs to reduce the pH to 6.80. What volume of 12.0 mol L⁻¹ hydrochloric acid does he need to add? Assume the volume of acid is insignificant compared to the volume of the pool.



Question 2: Dishwasher solution



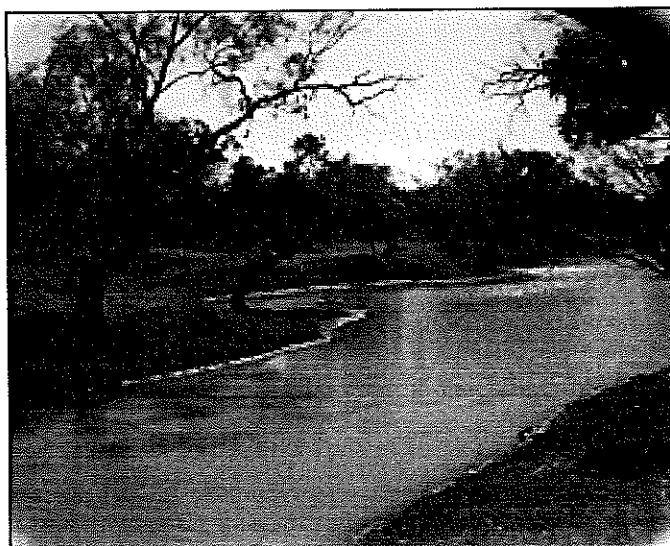
Question 7: Concrete batching plant

Notes

Set 16: The pH scale

Notes

10. (a) In an experiment to determine the effect of acid rain on the pH of a small lake near a lead smelter a researcher mixed equal volumes of water, one from the lake having a pH of 6.75 and the other from rain water collected during a winter downpour having a pH of 5.1. Calculate the pH of the resulting mixture. Assume that the solution volumes are additive.
- (b) In a different experiment designed to investigate methods of increasing the pH of the lake water the researcher mixed equal volumes of lake water and water of pH 8.00. Calculate the pH of the resulting solution. Again assume that the solution volumes are additive.

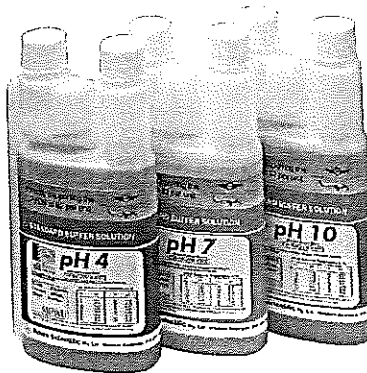


Question 10: Lake water

Set 17: Buffers

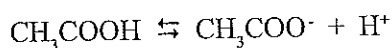
A buffer solution is one that resists changes in pH when small quantities of an acid or a base are added to it. They work by removing most of added hydrogen ions or hydroxide ions that would otherwise change the pH. Acidic and alkaline buffer solutions achieve this in different ways.

An acidic buffer solution has a pH less than 7. They are commonly made from a weak acid and one of its salts, a mixture of acetic acid and sodium acetate in solution for example. Solutions of the same concentration, mixed in equal molar proportions, will give the buffer a pH of around 4.76.



Buffer solution

Notes



The acetic acid and sodium acetate buffer solution contains the following active ingredients:

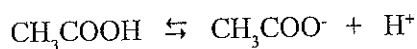
- a relatively high concentration of un-ionised acetic acid
- a relatively high concentration of acetate ions from the sodium acetate.
- a sufficient concentration of hydrogen ions to make the solution acidic.

Example

How will the acid buffer solution, acetic acid and sodium acetate, maintain its pH balance if

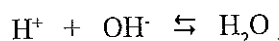
- extra acid (H^+) is added?
- extra alkali (OH^-) is added?

- To avoid the pH from dropping by the addition of extra hydrogen ions the buffer solution must remove them. It does this by combining the hydrogen ions with the acetate ions to make acetic acid. An equilibrium exists in the buffer solution, as illustrated by the equation.

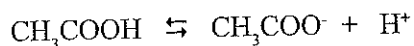


On the addition of extra H^+ the concentration of H^+ increases causing the equilibrium to shift towards the acetic acid and so using up most of the H^+ added. As a result the extra hydrogen ions are removed, so the pH does not change very much.

- To avoid the pH increasing by the addition of extra hydroxide ions the buffer solution must counter the change. When the extra hydroxide ions are added they react with the hydrogen ions of the buffer solution as illustrated by the equation.



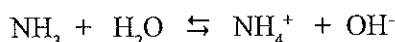
Removal of H^+ from the buffer on the addition of extra OH^- reduces the concentration of H^+ which in turn results in the equilibrium shifting towards the products replacing most of the H^+ used up by the added OH^- ions.



Set 17: Buffers

Notes

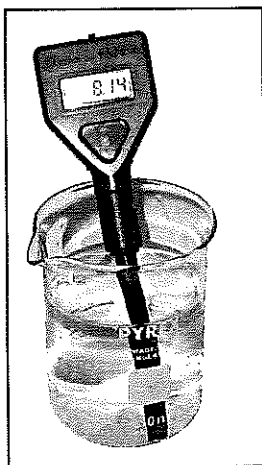
An alkaline buffer solution has a pH greater than 7. They are commonly made from a weak base and one of its salts. A mixture of ammonia solution and ammonium chloride solution is an example of an alkaline buffer. Solutions of the same concentration, mixed in equal molar proportions, will give the buffer a pH of 9.25.



The ammonia solution and ammonium chloride solution mixture gives a buffer with the following active ingredients:

- a relatively high concentration of unreacted ammonia.
- a relatively high concentration of ammonium ions from the ammonium chloride.
- a sufficient concentration of hydroxide ions to make the solution basic.

Set 17: Exercises



Question 3: pH meter

1. Describe the function of buffer solutions and comment on their importance in biological systems.
2. Explain how the alkaline buffer, ammonia solution and ammonium chloride solution mixture works to buffer against
 - (a) an increase in pH (addition of OH^-)
 - (b) a decrease in pH (addition of H^+)
3. A pH meter is a convenient tool for the measurement of the pH of a solution. It must however be calibrated regularly. Buffer solutions are used for the calibration process. Explain why buffer solutions are used for calibration of pH meters.
4.
 - (a) Why does a buffer need to be used in swimming pool water?
 - (b) Sodium hydrogencarbonate is added to swimming pool water for its buffering effect. Write an equation to help explain how it works.
5. Carbon dioxide dissolved in the blood helps to maintain a pH of 7.4.
 - (a) What other species need to be present for this process to work?
 - (b) Write equations to show how carbon dioxide is involved in maintaining this constant pH.
6.
 - (a) Starting with a 1 mol L^{-1} acetic acid solution, describe two methods of preparing a buffer solution.
 - (b) Write equations for any reactions that may occur.
7.
 - (a) Starting with sodium citrate ($\text{Na}_3\text{C}_6\text{H}_5\text{O}_7$) describe how you could prepare a buffer solution.
 - (b) Write equations for any reactions that may occur.
8. Commercial buffer solutions are usually supplied with temperature-pH charts that give the pH of the solution at various temperatures. Explain why the pH of a buffer solution varies with temperature.

Set 11: Periodic Trends

- The metallic character increases.
They have the same number of electrons.
The first ionisation energy decreases.
The electronegativity decreases.
The atomic radius increases.
- (a) This diagram tells us that as you move across the period from left to right, the atomic radii gets smaller despite the fact that there is an increase in the number of electrons present. This is due to the increase in the positive charge of the nucleus attracting the electrons more closely.
(b) As sodium loses an electron, the size of its ion is much smaller in diameter; the reverse is true for chlorine. The reason for this change is that as sodium loses the electron it also loses the only electron in the third energy level hence the size of the ion is smaller than the atom. The chlorine on the other hand as it gains the electron completes its third energy level and the size of its ion as a consequence is larger.
- (a) The electronegativity increases as you move to the right across a period since the number of protons in nucleus is increasing and electrons are attracted more closely.
(b) Electronegativity increases as you move up a group – same number of electrons but closer to nucleus. Fluorine is the most electronegative element.
- It could possibly be placed in group 2. It would tend to lose two electrons to form ions. As it has two valence electrons it is most likely a metallic element and would form ionic compounds. Metal properties eg: dense solid, high melting point, conduct electricity and heat, malleable & ductile.
- Bonding changes from metallic to covalent network to covalent molecular. Relating this to valence electrons:
 - 1-3 electrons Metallic
 - 3-4 electrons Covalent network
 - 5-7 electrons Covalent molecular

Acids and Bases

Set 12: Acid and Base strength

- Concentration is a measure of the proportion of solute in a solution.
Strength refers to the degree of ion formation that occurs when an acid or base is dissolved in water. For example, hydrogen chloride completely ionises in water and is classified as a strong acid:
 $\text{HCl(aq)} \rightarrow \text{H}^+(\text{aq}) + \text{Cl}^-(\text{aq})$
When the very soluble hydrogen fluoride dissolves in water there is little ionisation and it is classified as a weak acid: $\text{HF(aq)} \rightleftharpoons \text{H}^+(\text{aq}) + \text{F}^-(\text{aq})$
- (a) HCl , HNO_3 , H_2SO_4 , HBr , HI , HClO_4 ,
(b) CH_3COOH , H_2S , HF , H_3PO_4 , $\text{H}_2\text{C}_2\text{O}_4$, HSO_4^- , or any organic acid.
(c) NaOH , KOH , Ba(OH)_2 , Ca(OH)_2 , or any metal hydroxide or oxide.
(d) NH_3 , CO_3^{2-} , CH_3NH_2 or any organic amine.
- (a) Concentrated and weak (b) Dilute and strong (c) Concentrated and strong
(d) Dilute and weak
- (i) (a) $\text{HBr(g)} \rightarrow \text{H}^+(\text{aq}) + \text{Br}^-(\text{aq})$ (b) ionisation
(c) HBr , OH^- , $\text{Br}^- \approx \text{H}^+$ or H_3O^+ , H_2O
(ii) (a) $\text{CH}_3\text{COOH}(\ell) \rightleftharpoons \text{H}^+(\text{aq}) + \text{CH}_3\text{COO}^-(\text{aq})$ (b) ionisation
(c) OH^- , $\text{CH}_3\text{COO}^- \approx \text{H}^+$ or H_3O^+ , CH_3COOH , H_2O
(iii) (a) $\text{H}_2\text{SO}_4(\ell) \rightarrow \text{H}^+(\text{aq}) + \text{HSO}_4^-(\text{aq})$ and $\text{HSO}_4^-(\text{aq}) \rightleftharpoons \text{H}^+(\text{aq}) + \text{SO}_4^{2-}(\text{aq})$
(b) ionisation (c) H_2SO_4 , OH^- , SO_4^{2-} , HSO_4^- , H^+ or H_3O^+ , H_2O
(iv) (a) $\text{NH}_3(\text{g}) + \text{H}_2\text{O}(\ell) \rightleftharpoons \text{NH}_4^+(\text{aq}) + \text{OH}^-(\text{aq})$ (b) ionisation
(c) H_3O^+ or H^+ , $\text{NH}_4^+ \approx \text{OH}^-$, NH_3 , H_2O
(v) (a) $\text{Ba(OH)}_2(\text{aq}) \rightarrow \text{Ba}^{2+}(\text{aq}) + 2\text{OH}^-(\text{aq})$ (b) dissociation
(c) Ba(OH)_2 , H^+ or H_3O^+ , Ba^{2+} , OH^- , H_2O

Set 12: Acid and Base strength

5. Stronger acid is HClO_4 and the weaker base is ClO_4^-
6. (a) $\text{H}_2\text{S}(\text{g}) + \text{H}_2\text{O}(\ell) \rightleftharpoons \text{H}_3\text{O}^+(\text{aq}) + \text{HS}^-(\text{aq})$ and $\text{HS}^-(\text{g}) + \text{H}_2\text{O}(\ell) \rightleftharpoons \text{H}_3\text{O}^+(\text{aq}) + \text{S}^{2-}(\text{aq})$
 (b) Strongest acid is H_3O^+ and the strongest base is S^{2-}
7. (a) $\text{H}_3\text{PO}_4(\ell) \rightleftharpoons \text{H}^+(\text{aq}) + \text{H}_2\text{PO}_4^-(\text{aq})$
 $\text{H}_2\text{PO}_4^-(\text{aq}) \rightleftharpoons \text{H}^+(\text{aq}) + \text{HPO}_4^{2-}(\text{aq})$
 $\text{HPO}_4^{2-}(\text{aq}) \rightleftharpoons \text{H}^+(\text{aq}) + \text{PO}_4^{3-}(\text{aq})$
 (b) H_2O , H_3PO_4 , H^+ or H_3O^+ , H_2PO_4^- , HPO_4^{2-} , PO_4^{3-} , OH^-
8. 99% sulfuric acid consists mostly of H_2SO_4 molecules, as there is little water to allow ionisation, so there are very few hydrogen ions to react with the iron. 2 mol L^{-1} sulfuric acid is a strong acid; there is full ionisation for the release of one hydrogen ion represented by the equation:
 $\text{H}_2\text{SO}_4(\ell) \rightarrow \text{H}^+(\text{aq}) + \text{HSO}_4^-(\text{aq})$
 There is therefore a high concentration of hydrogen ions which is the species that reacts with the iron in a reaction represented by the equation $2\text{H}^+(\text{aq}) + \text{Fe}(\text{s}) \rightarrow \text{Fe}^{2+}(\text{aq}) + \text{H}_2(\text{g})$
9. For effective washing, large numbers of OH^- ions are required in the washing powder. To supply these substantial amounts of sodium hydroxide or sodium carbonate need to be present. The sodium hydroxide, being a strong base will produce a high concentration of OH^- ions which can be detrimental to fabrics and certainly to containers made from aluminium, zinc, chromium (a component of stainless steel) and any other amphoteric metal. It is also dangerous to use, as concentrated OH^- solutions are corrosive to skin. Sodium carbonate, however, is a weak base that produces relatively low concentrations of OH^- ions but as these are consumed, more are produced until all of the carbonate ions are used up. The production of the OH^- ions from the carbonate ions is illustrated by the equation: $\text{CO}_3^{2-}(\text{aq}) + \text{H}_2\text{O}(\ell) \rightleftharpoons \text{HCO}_3^-(\text{aq}) + \text{OH}^-(\text{aq})$
10. For this use the hydrogen ion concentration produced in the solution must be very low but there must be a process that replaces the hydrogen ions that are used up. It must therefore be a weak acid. As hydrochloric acid is a strong acid the concentration of hydrogen ions, even in dilute solutions is relatively high and would damage human skin particularly sensitive areas such as eyes and areas where fungal infections are likely to occur.

Set 13: Hydrolysis

1. (a) neutral (b) basic (c) basic
 (d) acidic (e) basic (f) acidic
3. (a) Ammonium nitrate, calcium hydrogenphosphate, potassium sulfate, and ammonium chloride.
 (b) The salts contain either an anion of a weak acid or the cation of a weak base.
 (c) Ammonium nitrate ($\downarrow \text{pH}$) $\text{NH}_4^+(\text{aq}) + \text{H}_2\text{O}(\ell) \rightleftharpoons \text{NH}_3(\text{aq}) + \text{H}_3\text{O}^+(\text{aq})$
 calcium hydrogenphosphate ($\uparrow \text{pH}$) $\text{HPO}_4^{2-}(\text{aq}) + \text{H}_2\text{O}(\ell) \rightleftharpoons \text{H}_2\text{PO}_4^-(\text{aq}) + \text{OH}^-(\text{aq})$
 and $\text{H}_2\text{PO}_4^-(\text{aq}) + \text{H}_2\text{O}(\ell) \rightleftharpoons \text{H}_3\text{PO}_4(\text{aq}) + \text{OH}^-(\text{aq})$
 potassium sulfate ($\uparrow \text{pH}$) $\text{SO}_4^{2-}(\text{aq}) + \text{H}_2\text{O}(\ell) \rightleftharpoons \text{HSO}_4^-(\text{aq}) + \text{OH}^-(\text{aq})$
 ammonium chloride ($\downarrow \text{pH}$) $\text{NH}_4^+(\text{aq}) + \text{H}_2\text{O}(\ell) \rightleftharpoons \text{NH}_3(\text{aq}) + \text{H}_3\text{O}^+(\text{aq})$
 (d) These salts contain only ions derived from strong acids and strong bases so they do not react with water to produce the strong acid or strong base.
4. $\text{Ca}(\text{H}_2\text{PO}_4)_2(\text{s}) \rightleftharpoons 2\text{H}_2\text{PO}_4^-(\text{aq}) + \text{Ca}^{2+}(\text{aq})$ and $\text{H}_2\text{PO}_4^-(\text{aq}) + \text{H}_2\text{O}(\ell) \rightleftharpoons \text{HPO}_4^{2-}(\text{aq}) + \text{H}_3\text{O}^+(\text{aq})$
5. (a) Yes, it becomes basic.
 (b) $\text{Na}_2\text{CO}_3(\text{aq}) \rightarrow 2\text{Na}^+(\text{aq}) + \text{CO}_3^{2-}(\text{aq})$ and $\text{CO}_3^{2-}(\text{aq}) + \text{H}_2\text{O}(\ell) \rightleftharpoons \text{HCO}_3^-(\text{aq}) + \text{OH}^-(\text{aq})$
6. (a) $\text{OH}^-(\text{aq}) + \text{CH}_3\text{COOH}(\text{aq}) \rightleftharpoons \text{H}_2\text{O}(\ell) + \text{CH}_3\text{COO}^-(\text{aq})$
 (b) The solution would be basic, $\text{pH} > 7$.
 (c) $\text{CH}_3\text{COO}^-(\text{aq}) + \text{H}_2\text{O}(\ell) \rightleftharpoons \text{CH}_3\text{COOH}(\text{aq}) + \text{OH}^-(\text{aq})$
7. Yes it becomes basic as F^- is the anion of a weak acid. $\text{F}^-(\text{aq}) + \text{H}_2\text{O}(\ell) \rightleftharpoons \text{HF}(\text{aq}) + \text{OH}^-(\text{aq})$
8. The pH of a solution of ammonium ethanoate depends on the relative strength of the weak base ammonia and the weak acid ethanoic acid. The hydrolysis process is represented by the equations:
 $\text{NH}_4^+(\text{aq}) + \text{H}_2\text{O}(\ell) \rightleftharpoons \text{NH}_3(\text{aq}) + \text{H}_3\text{O}^+(\text{aq})$ and
 $\text{CH}_3\text{COO}^-(\text{aq}) + \text{H}_2\text{O}(\ell) \rightleftharpoons \text{CH}_3\text{COOH}(\text{aq}) + \text{OH}^-(\text{aq})$
 Their ionisation constants are both close to 1.8×10^{-5} so the solution will be neutral.

Set 14: Water Equilibrium

- $\text{H}_2\text{O}(\ell) + \text{H}_2\text{O}(\ell) \rightleftharpoons \text{OH}^-(\text{aq}) + \text{H}_3\text{O}^+(\text{aq})$
- (a) $[\text{OH}^-] = 6.76 \times 10^{-8} \text{ mol L}^{-1}$ (b) Acidic
- $[\text{H}^+] = 6.33 \times 10^{-9} \text{ mol L}^{-1}$
- (a) $[\text{H}^+] = 6.45 \times 10^{-11} \text{ mol L}^{-1}$ (b) $[\text{H}^+] = 1.28 \times 10^{-13} \text{ mol L}^{-1}$
- (a) $[\text{OH}^-] = 9.62 \times 10^{-14} \text{ mol L}^{-1}$
 (b) $\text{H}_2\text{SO}_4(\ell) \rightarrow \text{H}^+(\text{aq}) + \text{HSO}_4^-(\text{aq})$
 and $\text{HSO}_4^-(\text{aq}) \rightarrow 0.9\text{HSO}_4^-(\text{aq}) + 0.1\text{H}^+(\text{aq}) + 0.1\text{SO}_4^{2-}(\text{aq})$
 Overall $\text{H}_2\text{SO}_4(\ell) \rightarrow 0.9\text{HSO}_4^-(\text{aq}) + 1.1\text{H}^+(\text{aq}) + 0.1\text{SO}_4^{2-}(\text{aq})$
 $[\text{H}^+] = 1.1[\text{H}_2\text{SO}_4] = 1.1 \times 0.125 = 0.1375 \text{ mol L}^{-1}$
 $[\text{OH}^-] = 7.27 \times 10^{-14} \text{ mol L}^{-1}$
- (a) Since the ionisation of water, $\text{heat} + \text{H}_2\text{O}(\ell) \rightleftharpoons \text{H}^+(\text{aq}) + \text{OH}^-(\text{aq})$ is endothermic (the reverse reaction, neutralisation is exothermic) a change in temperature will change the proportion of reactants and products and so change the equilibrium constant.
 (b) If the temperature is decreased the equilibrium position will move to restore some of the heat removed. To do this hydrogen and hydroxide ions must combine to produce water and heat, hence their concentration will become smaller.
 (c) If temperature is increased a greater concentration of hydrogen ions will result.
 (d) As the temperature is increased the rate of both the forward and reverse reactions will increase, but the endothermic forward reaction will increase more than the reverse reaction leading to an increase in the hydrogen ion concentration.
- (a) $[\text{OH}^-] = 6.45 \times 10^{-10} \text{ mol L}^{-1}$
 (b) Some release of SO_2 from the smelter with elevated levels of CO_2 dissolving in the rainwater.
 (c) $\text{SO}_2(\text{aq}) + \text{H}_2\text{O}(\ell) \rightleftharpoons \text{HSO}_3^-(\text{aq}) + \text{H}^+(\text{aq})$ and $\text{CO}_2(\text{aq}) + \text{H}_2\text{O}(\ell) \rightleftharpoons \text{HCO}_3^-(\text{aq}) + \text{H}^+(\text{aq})$
- $[\text{H}^+] = 8.07 \times 10^{-5} \text{ mol L}^{-1}$ $[\text{OH}^-] = 1.24 \times 10^{-10} \text{ mol L}^{-1}$
- (a) $[\text{H}^+] = 7.33 \times 10^{-6} \text{ mol L}^{-1}$ $[\text{OH}^-] = 1.36 \times 10^{-9} \text{ mol L}^{-1}$
 (b) Excess $\text{Al}(\text{OH})_3$ and $\text{Mg}(\text{OH})_2$ would remain undissolved (both insoluble in water) and the concentration of both hydrogen and hydroxide ions would therefore be $1.00 \times 10^{-7} \text{ mol L}^{-1}$.
- (a) $V(\text{Mg}(\text{OH})_2) = 4.86 \text{ mL}$
 (b) $[\text{H}^+] = 8.34 \times 10^{-3} \text{ mol L}^{-1}$ $[\text{OH}^-] = 1.20 \times 10^{-12} \text{ mol L}^{-1}$

Set 15: Indicators and their use

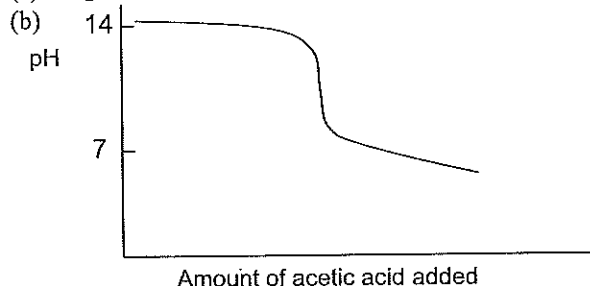
- (a) An organic acid or base
 (b) The acid must be distinctly different in colour to its conjugate base.
 E.g.: $\text{H}(\text{Litmus})(\text{aq}) \rightleftharpoons \text{H}^+(\text{aq}) + (\text{Litmus})^-(\text{aq})$

Red
Blue
- (a) $\text{H}(\text{Indicator})(\text{aq}) \rightleftharpoons \text{H}^+(\text{aq}) + (\text{Indicator})^-(\text{aq})$
 (b) Addition of an acid increases $[\text{H}^+]$, causes more $\text{H}(\text{Indicator})$ to be produced and a reduction in the concentration of $(\text{Indicator})^-$. The colour changes from the $(\text{Indicator})^-$ colour to the $\text{H}(\text{Indicator})$ colour.
 (c) Addition of a base reduces $[\text{H}^+]$ as H^+ reacts with OH^- to produce water. This causes the production of more $(\text{Indicator})^-$ and less $\text{H}(\text{Indicator})$. The colour changes from $\text{H}(\text{Indicator})$ colour to $(\text{Indicator})^-$ colour.
 (d) *Addition of Acid:* increase in the rate of the reaction producing $\text{H}(\text{Indicator})$ with no immediate change to the reaction producing $(\text{Indicator})^-$ hence a colour change from $(\text{Indicator})^-$ colour to $\text{H}(\text{Indicator})$ colour.
Addition of Base: a reduction in $[\text{H}^+]$ causes a reduction in the rate of the reaction producing $\text{H}(\text{Indicator})$ with no immediate change in the rate of the reaction producing $(\text{Indicator})^-$ hence a colour change from $\text{H}(\text{Indicator})$ colour to $(\text{Indicator})^-$ colour.

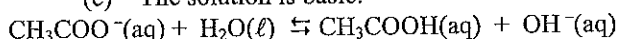
(a)	Phenolphthalein	Alizarin Yellow	Indigo Carmine
(b)	$<\text{pH} = 8.3$ (Colourless) $>\text{pH} = 10$ (Pink)	$<\text{pH} = 10.1$ (Yellow) $>\text{pH} = 12.0$ (Red)	$<\text{pH} = 11.4$ (Blue) $>\text{pH} = 13.0$ (Yellow)
(c)	8.3 to 10	10.1 to 12.0	11.4 to 13.0

4.

(a)	Methyl red	Methyl orange	Bromophenol blue	Methyl violet
(b)	<pH = 4.4 (Red) >pH = 6.2 (Yellow)	<pH = 3.1 (Red) >pH = 4.4 (Yellow)	<pH = 3.0 (Yellow) >pH = 4.6 (blue)	<pH = 0 (Yellow) >pH = 1.6 (Violet)
(c)	4.4 to 6.2	3.1 to 4.4	3.0 to 4.6	0 to 1.6

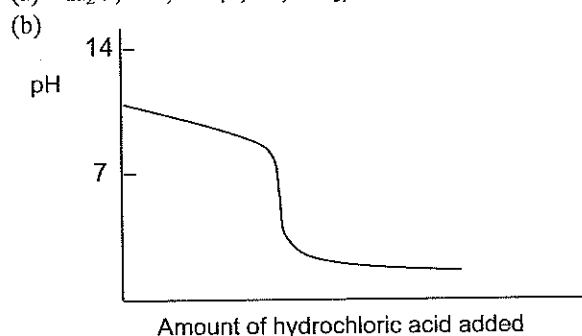
5. (a) H_2O Na^+ CH_3COO^- OH^- CH_3COOH H^+ 

(c) The solution is basic.

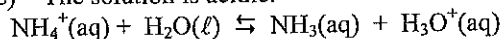


(d) Phenolphthalein.

The end point (the point where the colour changes) must occur at the equivalence point. As the equivalence point is basic an indicator that changes colour at a pH between about 7 and 11 is required. Phenolphthalein changes colour in the pH range 8.3 to 10.

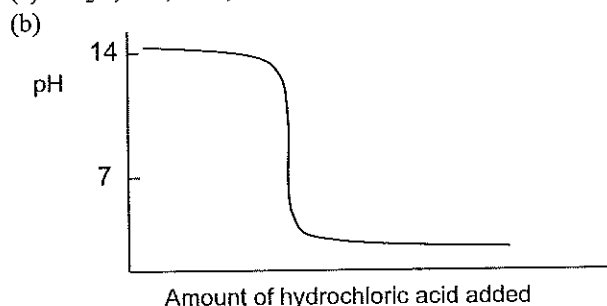
6. (a) H_2O , Cl^- , NH_4^+ , H^+ , NH_3 , OH^- 

(c) The solution is acidic.



(d) methyl orange, methyl red or bromothymol blue.

The end point (the point where the colour changes) must occur at the equivalence point. As the equivalence point is acidic an indicator that changes colour at a pH between about 3 and 7 is required. (see answer to Q4)

7. (a) H_2O , Cl^- , Ca^{2+} , H^+ $\text{OH}^- = 1.0 \times 10^{-7} \text{ mol L}^{-1}$ 

(c) The solution is neutral. Neither the calcium ion nor the chloride ion will react with water to undergo hydrolysis.

(d) Most indicators will give the correct end point which is at pH = 7 as long as they change colour somewhere in the range of pH = about 3 to 11.

8. (a) Phenolphthalein. (b) More acid will be used.
(c) The concentration will be measured lower than it actually is.

Set 16: The pH scale

- $[\text{H}^+] = [\text{HCl}] = 0.100 \text{ mol L}^{-1}$ $[\text{OH}^-] = 1.00 \times 10^{-13} \text{ mol L}^{-1}$; pH = 1.00
 - $[\text{H}^+] = [\text{HNO}_3] = 0.00500 \text{ mol L}^{-1}$ $[\text{OH}^-] = 2.00 \times 10^{-12} \text{ mol L}^{-1}$; pH = 2.30
 - $[\text{OH}^-] = [\text{NaOH}] = 0.0100 \text{ mol L}^{-1}$ $[\text{H}^+] = 1.00 \times 10^{-12} \text{ mol L}^{-1}$; pH = 12.0
 - $[\text{H}^+] = [\text{HCl}] = 2.00 \text{ mol L}^{-1}$ $[\text{OH}^-] = 5.00 \times 10^{-15} \text{ mol L}^{-1}$; pH = 0.300
 - $[\text{H}^+] = [\text{OH}^-] = 1.00 \times 10^{-14} \text{ mol L}^{-1}$ and the pH = 7.00. Neutral as solutions of Na^+ ions and Cl^- ions do not hydrolyse in water.
- $[\text{H}^+] = 1.00 \times 10^{-3} \text{ mol L}^{-1}$ $[\text{OH}^-] = 1.00 \times 10^{-11} \text{ mol L}^{-1}$
 - $[\text{H}^+] = 1.00 \times 10^{-11} \text{ mol L}^{-1}$ $[\text{OH}^-] = 1.00 \times 10^{-3} \text{ mol L}^{-1}$
 - $[\text{H}^+] = 10.0 \text{ mol L}^{-1}$ $[\text{OH}^-] = 1.00 \times 10^{-15} \text{ mol L}^{-1}$
 - $[\text{H}^+] = 2.75 \times 10^{-5} \text{ mol L}^{-1}$ $[\text{OH}^-] = 3.63 \times 10^{-10} \text{ mol L}^{-1}$
 - $[\text{H}^+] = 2.51 \times 10^{-8} \text{ mol L}^{-1}$ $[\text{OH}^-] = 3.98 \times 10^{-7} \text{ mol L}^{-1}$

3. Concentration changed by a factor of 1000
4. $[\text{H}^+] = 1.80 \text{ mol L}^{-1}$
5. Water required = $628 - 25.0 = 603 \text{ mL}$
6. $m(\text{NaOH})_{\text{to be added}} = nM = 4.99 \times 10^{-4} \times 39.998 = 0.00200 \text{ g} = 20.0 \text{ mg}$
7. (a) $\text{pH} = -\log_{10}[\text{H}^+] = -\log_{10}(5.32 \times 10^{-14}) = 13.3$
(b) $V(\text{HCl}) = n(\text{H}^+) \div c = 0.188 \text{ L} = 188 \text{ mL}$
8. $\text{pH} = -\log_{10}[\text{H}^+] = -\log_{10}(7.40 \times 10^{-10}) = 9.13$
9. $n(\text{HCl})_{\text{required}} = 2.853 \text{ mol}$
 $V(\text{HCl}) = n(\text{H}^+)_{\text{required}} \div c = 23.8 \text{ mL}$
10. (a) $\text{pH} = -\log_{10}[\text{H}^+] = -\log_{10}(4.059 \times 10^{-6}) = 5.39$
(b) $n(\text{OH}^-)_{\text{excess}} = n(\text{OH}^-)_{\text{at pH } 8.00} - n(\text{H}^+)_{\text{at pH } 6.75}$
 $\text{pH} = -\log_{10}[\text{H}^+] = -\log_{10}(2.43 \times 10^{-8}) = 7.61$

Set 17: Buffers

1. Buffer solutions resist a change in pH even with the addition of substantial amounts of hydrogen or hydroxide ions. Many specific reactions that occur in biological systems occur only at specific pH values. Some reactions produce or use up hydrogen ions in these solutions. Buffers prevent large changes in the pH of solutions such as blood, cell contents and lymph system allowing vital reactions to continue.
3. Buffer solutions can be produced that have specific and known pH value. These values remain constant (at constant temperature) even with the inadvertent addition of small amounts of H^+ or OH^- ions. The fixed pH values are used as standards to calibrate the meters.
4. (a) Hypochlorite ions are added to pool water in the form of sodium hypochlorite solution or calcium hypochlorite solid to kill micro organisms. Some of the hypochlorite ions react with water to produce the weak acid hypochlorous acid hydroxide ions as illustrated in the equation

$$\text{OCl}^-(\text{aq}) + \text{H}_2\text{O}(\ell) \rightleftharpoons \text{HOCl}(\text{aq}) + \text{OH}^-(\text{aq})$$

The buffer uses up some of these OH^- ions stopping the pool water becoming alkaline too quickly.
- (b) $\text{OH}^-(\text{aq}) + \text{HCO}_3^-(\text{aq}) \rightleftharpoons \text{H}_2\text{O}(\ell) + \text{CO}_3^{2-}(\text{aq})$ and $\text{HCO}_3^-(\text{aq}) \rightleftharpoons \text{CO}_3^{2-}(\text{aq}) + \text{H}^+(\text{aq})$ where the H^+ reacts with the OH^- . Both processes use up OH^- ions.
5. (a) H_2O and HCO_3^- ion
(b) $\text{CO}_2(\text{aq}) + \text{H}_2\text{O}(\ell) \rightleftharpoons \text{HCO}_3^-(\text{aq}) + \text{H}^+(\text{aq})$
 H^+ ion is used up by reacting with the HCO_3^- as indicated in the reverse reaction above and OH^- ions is used up by reacting with CO_2 directly, $\text{OH}^-(\text{aq}) + \text{CO}_2(\text{aq}) \rightleftharpoons \text{HCO}_3^-(\text{aq})$
or with H^+ ions which results in more H^+ ions being produced. $\text{H}^+(\text{aq}) + \text{OH}^-(\text{aq}) \rightleftharpoons \text{H}_2\text{O}(\ell)$
6. Method 1: Add an equal volume of 1 mol L^{-1} sodium ethanoate solution
Method 2: Add an equal volume of a 0.5 mol L^{-1} solution of sodium hydroxide. This would react with half the acetic acid to produce acetate ion as in the equation

$$\text{CH}_3\text{COOH}(\text{aq}) + \text{OH}^-(\text{aq}) \rightleftharpoons \text{CH}_3\text{COO}^-(\text{aq}) + \text{H}_2\text{O}(\ell)$$
7. Method 1: Make a solution in water of the sodium citrate mixed with an equal number of moles of citric acid.
Method 2: Make a solution in water of the sodium citrate then add $1\frac{1}{2}$ times that number of moles of hydrochloric acid. This would react with half the citrate ion to produce citric acid as in the equation: $\text{C}_6\text{H}_5\text{O}_7^{3-}(\text{aq}) + 3\text{H}^+(\text{aq}) \rightleftharpoons \text{H}_3\text{C}_6\text{H}_5\text{O}_7(\text{aq})$
8. A change in temperature changes the equilibrium concentration of the species in equilibrium and as one of the species is always either H^+ or OH^- ions, changing their concentrations changes the pH.